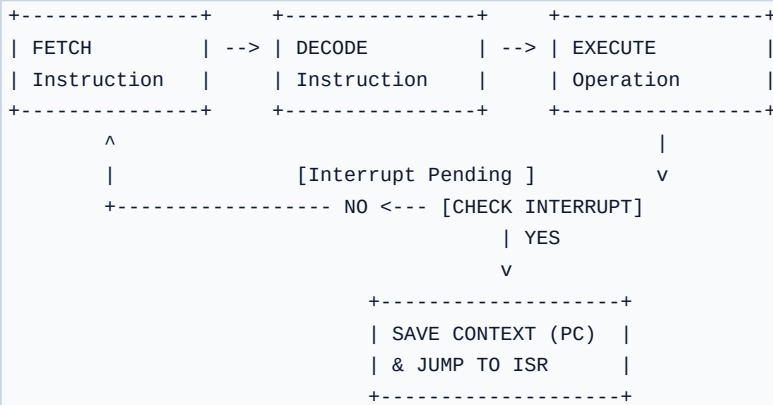


3-Mark Questions & Answers -- MPCA Unit 2: Processor Organization & Architecture

Q1. Explain the basic instruction cycle with interrupt processing.



* Fetch: Load instruction from memory (PC -> MAR -> RAM -> IR).

* De

Q2. Differentiate between hardwired and microprogrammed control unit.

Feature	Hardwired Control Unit	Microprogrammed Control Unit
Implementation	Fixed logic gates, flip-flops, decoder	Microcode software stored in Control M
Speed	Very Fast (hardware gate speeds).	Slower (requires ROM access cycles).
Flexibility / Modifiability	Rigid; hard to modify or update.	Flexible; easy to update microcode.
Cost & Design	Complex design, costly for complex ISA	Systematic design, cheaper for CISC.

Q3. Explain types of pipeline hazards (structural, data, control) briefly.

1. Structural Hazard (Resource Conflict): Occurs when two instructions in different stages attempt to access the same physical hardware resource simultaneously (e.g., single memory port for instruction fetch and data read).

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Q4. Explain Flynn's classification (SISD, SIMD, MISD, MIMD) with examples.

1. SISD (Single Instruction, Single Data): Classic single-core processor executing one stream of scalar data (e.g., Intel 8086, traditional single-core CPUs).

2. S
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Q5. What are micro-operations Give examples.

Micro-operations are elementary functional operations executed by CPU hardware during a single clock pulse to perform register transfers, ALU calculations, or bus reads.

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